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Experiment-3: Decision Tree

Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

Python Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree

# Read CSV file
data = pd.read_csv("play_tennis.csv")

# Encode categorical values
encoders = {}

for col in data.columns:
    le = LabelEncoder()
    data[col] = le.fit_transform(data[col])
    encoders[col] = le

# Features and Target
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
```

```

# Train ID3 Decision Tree
model = DecisionTreeClassifier(criterion="entropy", random_state=0)
model.fit(X, y)

# Plot only the decision tree
plt.figure(figsize=(14, 8))

plot_tree(
    model,
    feature_names=["Outlook", "Temperature", "Humidity", "Wind"],
    class_names=["No", "Yes"],
    filled=True,
    rounded=True,
    fontsize=12
)

# Title
plt.title("Decision Tree using ID3 Algorithm", fontsize=18, fontweight="bold")

# Hide axes
plt.axis("off")

# Show tree
plt.show()

```

playtennis.csv

```

Outlook,Temperature,Humidity,Wind,PlayTennis
Sunny,Hot,High,Weak,No
Sunny,Hot,High,Strong,No
Overcast,Hot,High,Weak,Yes

```

Rain,Mild,High,Weak,Yes

Rain,Cool,Normal,Weak,Yes

Rain,Cool,Normal,Strong,No

Overcast,Cool,Normal,Strong,Yes

Sunny,Mild,High,Weak,No

Sunny,Cool,Normal,Weak,Yes

Rain,Mild,Normal,Weak,Yes

Sunny,Mild,Normal,Strong,Yes

Overcast,Mild,High,Strong,Yes

Overcast,Hot,Normal,Weak,Yes

Rain,Mild,High,Strong,No

Output:

Decision Tree using ID3 Algorithm

